Engineering Math's by Prof. Nitin

Practice Exercise for Sem II

Topic : ***Module 1: Differential equations of First order and First degree***

1 Solve $\frac{dy}{dx} = \frac{tany - 2xy - y}{x^2 - xtan^2 y + \sec^2 y}$

2 Solve $(1+y^2)dx = \left(\sqrt{1+y^2}\sin y - xy\right)dy$

3 Solve $ydx - xdy + \log x\, dx = 0$

4 Solve $y(2xy+1)dx + x(1+2xy-x^3y^3)dy = 0$

5 Solve $xe^x(dx-dy) + e^x dx + ye^y dy = 0$

6 Solve $\frac{dy}{dx} - xy = y^2 e^{-\frac{x^2}{2}} \log x$

7 Solve $(\mathrm{x}^4 + \mathrm{y}^4)\mathrm{dx} - \mathrm{xy}^3\mathrm{dy} = 0$

8 Solve $(\mathrm{y}\log \mathrm{y})\mathrm{dx} + (\mathrm{x} - \log \mathrm{y})\mathrm{dy} = 0$

9 Solve $\frac{\mathrm{dy}}{\mathrm{dx}} = e^{x-y}(e^x - e^y)$

10 Solve $(1+y^2)dx = (e^{\tan^{-1} y} - x)dy$

11 Solve $\frac{dy}{dx} - \frac{\tan y}{1+x} = (1+x)e^x \sec y$

12 Solve $(1+\sin y)\frac{dy}{dx} = [2y\cos y - x(\sec y + \tan y)]$

13 Solve $\frac{dz}{dx} + \frac{z}{x}\log z = \frac{z}{x^2}(\log z)^2$

14 Solve $\frac{dy}{dx} + x(x+y) = x^3(x+y)^3 - 1$

15 Solve $(xy\sin xy + \cos xy)ydx + (xy\sin xy - \cos xy)xdy = 0$

16 Solve $y(x+y)dx - x(y-x)dy = 0$

17 Solve $y(xy+e^x)dx - e^x dy = 0$

18 Solve $(e^y+1)\cos x\, dx + e^y \sin x\, dy = 0$

19 Solve $x\sin x\, dy + (xy\cos x - y\sin x - 2)dx = 0$

20 Solve $(y - xy^2)dx - (x + x^2y)dy = 0$

Topic : ***Module 2: Linear differential equations with constant coefficients and variable coefficients of higher order***

1 Solve $(D^3 - 5D^2 + 8D - 4)y = 0$

2 Solve $(D^4 + 6D^2 + 9)y = 0$

3 Solve $(D^3 + 3D)y = \cosh 2x \sinh 3x$

4 Solve $(D^3 - 2D + 4)y = 3x^2 - 5x + 2$

5 Solve $(D^3 + D)y = \cos t + t^2 + 3$

6 Solve $(D - 1)^2(D^2 + 1)y = 3x^2 - 5x + 2$

7 Solve $(D^2 - 8D + 16)y = \frac{e^{4x}}{x^2}$

8 Solve $(D^2 - 4D + 4)y = e^{2x} + x^3 + \cos 2x$

9 Solve $(D^2 + 4D + 4)y = xe^{-x} \sinh x$

10 Solve $(\mathrm{D}^2 + 2\mathrm{D} + 1)\mathrm{y} = \mathrm{xe}^{-\mathrm{x}} \cos \mathrm{x}$

11 Solve $(\mathrm{D}^2 - 4\mathrm{D} + 4)\mathrm{y} = \mathrm{e}^{2\mathrm{x}} \sec^2 \mathrm{x}$

12 Solve $(\mathrm{D}^2 - 1)\mathrm{y} = \mathrm{x} \sin 3x + \cos x$

13 Solve $(\mathrm{D}^4 + 2\mathrm{D}^2 + 1)\mathrm{y} = \mathrm{x}^2 \cos x$

14 Solve $(D^2 - 1)y = x^2 \cos 2x$

15 Solve by method of variation of parameters, $\frac{d^2y}{dx^2} + y = \sec x \tan x$

16 Solve by method of variation of parameters, $(D^2 - 6D + 9)y = \frac{e^{3x}}{x^2}$

17 Solve by method of variation of parameters, $(D^2 - 4D + 4)y = e^{2x} \sec^2 x$

18 Solve by method of variation of parameters, $(D^2 - 1)y = \frac{2}{1+e^x}$

19 Solve by method of variation of parameters, $(D^2 + 2D + 1)y = e^{-x} \log x$

20 Solve by method of variation of parameters $(D^2 + 3D + 2)y = \frac{1}{1+e^x}$

Topic : ***Module 3: Beta Gamma Functions***

1 Show that $\int_0^\infty xe^{-x^8} dx \int_0^\infty x^2\, e^{-x^4} dx = \frac{\pi}{16\sqrt{2}}$

2 Prove that $\int_0^1 \sqrt{1 - \sqrt{x}}\, dx \int_0^{1/2} \sqrt{2y - 4y^2}\, dy = \frac{\pi}{30}$

3 Evaluate $\int_0^1 x^5 \sin^{-1} x\, dx$

4 Evaluate $\int_0^\infty \frac{x^2}{(1+x^6)^{7/2}}\, dx$

5 Evaluate $\int_0^\infty x^2 7^{-4x^2} dx$

6 Evaluate $\int_0^1 x^3 \left(\log \frac{1}{x}\right)^4 dx$

7 Evaluate $\int_0^{\pi/4} \cos^3 2\theta \sin^2 4\theta \; d\theta$

8 Prove that $\int_3^7 \sqrt[4]{(x - 3)(7 - x)}dx = \frac{2\left(\gamma\left(\frac{1}{4}\right)\right)^2}{3\sqrt{\pi}}$

9 Prove that $\beta(x,x)=\frac{1}{2^{2x-1}}\beta\left(x,\frac{1}{2}\right)$

10 Prove that $\beta(m,m).\beta\left(m+\frac{1}{2},m+\frac{1}{2}\right)=\frac{\pi}{m}.2^{1-4m}$

11 Prove that $\beta(x,x)=\frac{1}{2^{2x-1}}\ \beta\left(x,\frac{1}{2}\right)$

12 Prove that $\int_0^{2\pi}\sin^2\theta\,(1+\cos\theta)^4 d\theta=\frac{21\pi}{8}$

13 Prove that $\int_{-\pi/2}^{\pi/2}\cos^3\theta\,(1+\sin\theta)^2 d\theta=\frac{8}{5}$

14 Prove that $\int_0^{\infty}\frac{\mathrm{dx}}{(\mathrm{e^x+e^{-x}})^{\mathrm{n}}}=\frac{1}{4}\ \beta\left(\frac{\mathrm{n}}{2},\frac{\mathrm{n}}{2}\right)$

15 Assuming the validity of differentiation under the integral sign prove that $\int_0^{\infty}\frac{e^{-\alpha x}\sin x}{x}dx=\cot^{-1}x.$

16 Assuming the validity of differentiation under the integral sign prove that $\int_0^{\infty}\frac{\log(1+ax^2)}{x^2}dx=\pi\sqrt{a}$

17 Assuming the validity of differentiation under the integral sign prove that $\int_0^{\infty}xe^{-ax}\sin bx\,dx=\frac{2ab}{(a^2+b^2)^2}.$

18 By the rule of differential under integral sign Evaluate $\int_0^{\pi/2}\frac{\mathrm{dx}}{\mathrm{a^2sin^2x+b^2cos^2x}}$

19 Find the total length of the curve $\mathrm{x}^{2/3}+\mathrm{y}^{2/3}=\mathrm{a}^{2/3}$

20 Find the length of the arc of the curve $y=\log\sec x$ from $x=0$ to $x=\pi/3$.

Topic : ***Module 4: Multiple Integrals – I***

1 Evaluate $\int_0^1\int_{x^2}^{x}xy(x^2+y^2)dydx$

2 Evaluate $\int_0^1\int_0^{\sqrt{1+x^2}}\frac{1}{1+x^2+y^2}dxdy$

3 Evaluate $\int_0^{\pi/4}\int_0^{\sqrt{\cos 2\theta}}\frac{r}{(1+r^2)^2}drd\theta$

4 Change the order of integration and evaluate $\int_0^2\int_{2-\sqrt{4-y^2}}^{2+\sqrt{4-y^2}}dx\,dy$

5 Change the order of integration and evaluate $\int_0^a\int_0^{a-\sqrt{a^2-y^2}}\ \frac{xy\log(x+a)}{(x-a)^2}dx\,dy$

6 Change the order of integration and evaluate $\int_0^a\int_{x^2/a}^{2a-x}\ xy\,dx\,dy$

7 Change the order of integration by expressing it as a single integral and evaluate $\int_0^1\int_0^y\ (x^2+y^2)\,dx\,dy+\int_1^2\int_0^{2-y}\ (x^2+y^2)\,dx\,dy$

8 Change the order of integration and evaluate $\int_0^1\int_x^{\sqrt{2-x^2}}\frac{x}{\sqrt{x^2+y^2}}dx\,dy$

9 Evaluate $\iint\sin\theta\,dA$ over R, where R is the region in the first quadrant that is outside the circle $r=2$ and inside the cardioid $r=2(1+\cos\theta\,)$

10 Evaluate $\iint\frac{r}{\sqrt{r^2+4}}\ drd\theta$ over one loop of $r^2=4\cos 2\theta$.

11 Evaluate $\iint xy\,dxdy$ over the region R given by $x^2+y^2-2x=0$, $y^2=2x$ and $y=x$

12 Evaluate $\iint x(x-y)\,dxdy$ over the region R which is triangle whose vertices are $(0,0),(1,2),(0,4)$

13 Evaluate $\iint x^2\,dxdy$ over the region R in the first quadrant bounded by $xy=16, y=x, y=0$ and $x=8$.

14 Change the order of integration and evaluate $\int_0^2 \int_{\sqrt{2x}}^{2} \frac{y^2}{\sqrt{y^4-4x^2}}\,dx\,dy$

15 Change the order of integration and evaluate $\int_0^2 \int_0^{4-x^2} \frac{xe^{2y}}{4-y}\,dx\,dy$

16 Change the order of integration and evaluate $\int_0^3 \int_{y^2/9}^{\sqrt{10-y^2}}\,dx\,dy$

17 Evaluate $\iint r^3 dr\,d\theta$ over the area bounded between the circles $r=2\sin\theta$ and $r=4\sin\theta$

18 Evaluate $\iint \frac{y}{(a-x)\sqrt{ax-y^2}}\,dxdy$ over the region R bounded by $y^2=ax$ and $y=x$.

19 Evaluate $\int_0^{a\sqrt{3}} \int_0^{\sqrt{x^2+a^2}} \frac{x}{y^2+x^2+a^2}\,dydx$

20 Evaluate $\int_{-\pi/4}^{\pi/4} \int_0^{\sqrt{\cos 2\theta}} \frac{r}{(1+r^2)^2}\,drd\theta$

Topic : ***Module 5: Multiple Integrals – II***

1 Find by double integration the area common to the circles $x^2+y^2=10$ and the parabola $y^2=9x$.

2 Find by double integration the area between the parabola $y=x^2-6x+3$ and the line $y=2x-9$.

3 Find by double integration the area common to the circles $x^2+y^2-4y=0$ and $x^2+y^2-4x-4y+4=0$.

4 Find by double integration the area between the curves $y^2=4x$ and $2x-3y+4=0$.

5 Find by double integration the area of the smaller region bounded by $x^2+y^2=a^2$ and $x+y=a$.

6 Find the area inside the circle $r=a\sin\theta$ and outside the cardioide $r=a(1-\cos\theta)$.

7 Find the area common to the circles $r=a$ and $r=2a\cos\theta$.

8 Find the area common to the cardioides $r=(1-\cos\theta)$ and $r=(1+\cos\theta)$.

9 Find the area of upper half of the cardioide $r=a(1+\cos\theta)$.

10 Find the area common to the circles $r=a\cos\theta$ and $r=a\sin\theta$.

11 Evaluate $\int_{-1}^{1} \int_0^{z} \int_{x-z}^{x+z} (x+y+z)\,dydxdz$

12 Evaluate $\int_0^{\pi/2} \int_0^{a\sin\theta} \int_0^{(a^2-r^2)/a} r\,dzdrd\theta$

13 Evaluate $\int_0^2 \int_0^x \int_0^{2x+2y} e^{(x+y+z)}\,dzdydx$

14 Evaluate $\iiint xyz\,dzdydx$ throughout the volume bounded by $x=0, y=0, z=0$, and $x+y+z=1$.

15 Evaluate $\iiint x^2yz\,dzdydx$ throughout the volume bounded by $x=0, y=0, z=0$, and $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$.

16 Evaluate $\iiint x^2+y^2+z^2\,dzdydx$ over the first octant of the sphere $x^2+y^2+z^2=a^2$.

17 Evaluate $\iiint \frac{z^2}{x^2+y^2+z^2}\,dzdydx$ over the volume of the sphere $x^2+y^2+z^2=2$.

18 Evaluate $\iiint z^2\,dxdydz$ over the volume common to the sphere $x^2+y^2+z^2=a^2$ and the cylinder $x^2+y^2=ax$.

19 Evaluate $\iiint z^2 dxdydz$ over the volume bounded by the cylinder $x^2 + y^2 = a^2$, the paraboloid $x^2 + y^2 = z$ and the plane $z = 0$.

20 Evaluate $\iiint \sqrt{x^2 + y^2} dzdydx$ over the volume bounded by the right circular cone $x^2 + y^2 = z^2, z > 0$ and the planes $z = 0$ and $z = 1$.

Topic : ***Module 6: Numerical Methods***

1 Using Euler's method find the approximate value of y when $\frac{dy}{dx} = x^2 + y^2$ and $y(0) = 1$, at $x = 1$ in five steps.

2 Using Euler's method find the approximate value of y when $\frac{dy}{dx} = \frac{y-x}{\sqrt{xy}}$ and $y(1) = 2,$ at $x = 1.5$ in five steps taking $h = 0.1$.

3 Using Euler's method find the approximate value of y when $\frac{dy}{dx} = y^2 - \frac{y}{x}$ and $y(1) = 1,$ at $x = 1.6$ taking $h = 0.1$.

4 Using Euler's modified method find the approximate value of y when $\frac{dy}{dx} = \log_e(x + y)$ and $y(1) = 2,$ at $x = 1.2$ taking $h = 0.2$.

5 Using Runge-Kutta method of 2nd order find the value of y satisfying the equation $\frac{dy}{dx} = \log(x + y), y(1) = 2$, for $x = 1.2$ at $h = 0.2$.

6 Using Euler's Modified method find the value of y satisfying the equation $\frac{dy}{dx} = 2 + \sqrt{xy}$, $y(1.2) = 1.6403$, for $x = 1.6$ correct up to 4 decimal places by taking $h = 0.2$.

7 Using Euler's Modified method find the value of y satisfying the equation $\frac{dy}{dx} = x + 3y$, $y(0) = 1$, for $x = 0.2$ correct up to 4 decimal places by taking $h = 0.1$.

8 Using Runge - Kutta method of 4th order find the value of y satisfying the equation $\frac{dy}{dx} = \frac{(y^2-x^2)}{(y^2+x^2)}$, $y(0) = 1$, for $x = 0.2$ and $x = 0.4$

9 Using Runge - Kutta method of 4th order find the value of y satisfying the equation $\frac{dy}{dx} = \frac{1}{x+y}$, $y(0) = 1$, for $x = 0.5$ at $h = 0.5$.

10 Using Runge - Kutta method of 4th order find the value of y satisfying the equation $\frac{dy}{dx} = \frac{(y^2-x^2)}{(y^2+x^2)}$, $y(0) = 1$, for $x = 0.4$ taking $h = 0.2$.

11 Using Runge-Kutta method of 2nd order find the value of y satisfying the equation $\frac{dy}{dx} = x - y^2$, $y(0) = 1$, for $x = 0.1$ correct upto 4 places by taking $h = 0.05$.

12 Given the following values of e^x, evaluate $\int_0^{2.5} e^x \, dx$, using Trapezoidal rule.

X	0	0.5	1	1.5	2	2.5
$y = e^x$	1	1.65	2.72	4.48	7.39	12.18

13 A rocket is launched from the ground. Its acceleration is registered during the first 80 seconds and is given in the following table.

t	0	10	20	30	40	50	60	70	80
$a\left(\frac{m}{s^2}\right)$	30	31.63	33.34	35.47	37.75	40.33	43.25	46.69	50.67

Find the velocity of the rocket at $t = 80\ secs$, by 1) Trapezoidal rule, 2) Simpsons 1/3rd rule.

14 Find the value of $\int_0^1 \frac{x^2}{1+x^3} dx$ by dividing the interval in 6 equal parts using i) Trapezoidal Rule ii) Simpson's 1/3rd Rule and compare with exact value.

15 Find the value of $\int_0^{1.2} \frac{1}{1+x^2} dx$ by dividing the interval in 6 equal parts using i) Trapezoidal Rule ii) Simpson's 1/3rd Rule.

16 Find the value of $\int_4^{4.2} \log x\, dx$ by dividing the interval in 6 equal parts using Simpson's 1/3rd Rule.

17 Compute the value of $\int_0^{\pi/2} \frac{(\sin x)}{x} dx$ by using Simpson's 3/8th Rule by dividing the interval in to 6 equal parts.

18 Using Simpson's 3/8th rule find $\int_0^{0.3} \sqrt{1-8x^3}\ dx$ by dividing the interval in 3 equal parts.

19 Evaluate $\int_0^{\pi/2} \sqrt{\sin x + \cos x}\, dx$ by using Simpson's 3/8th Rule by dividing the interval in to 6 equal parts.

20 Evaluate $\int_{0.2}^{1.4} (\sin x - \log x + e^x) dx$ by using Simpson's 3/8th Rule by dividing the interval in to 6 equal parts.